

Feature Gap Analysis: EnergyV1 Project

Executive Summary

- **Total Features Analyzed:** 18
- **Complete Features:** 15
- **Partial Features:** 3
- **Missing Features:** 0
- **Regressions:** 0

Overall Completion Percentage: ~83% (15/18 fully complete, 3 partial).

Top Priority Improvements

1. **Filter Produk (UX Improvement):** Implement a comprehensive filtering sidebar (by price range, brand, attributes) instead of just the current sorting dropdown.
2. **Sistem PLTS 3 Page:** Create dedicated informative landing pages for On-Grid, Off-Grid, and Hybrid PLTS systems, rather than just treating them as product subcategories.
3. **Dedicated Contact Page:** Extract contact details from "Tentang Kami" into a dedicated `/kontak` page with a direct contact form and embedded map for better UX.

Summary Table

Feature	Current	Desired	Status	Priority	Complexity
Landing Page	Exists via <code>app/page.tsx</code>	Dedicated landing page	✅ Complete	Low	Low
Cart	Exists via <code>app/keranjang</code>	Working Cart	✅ Complete	Low	Low
All katalog Product	Exists via <code>app/produk</code>	All products catalog	✅ Complete	Low	Low
Katalog per masing masing	Exists via <code>app/kategori/[slug]</code>	Category specific catalog	✅ Complete	Low	Low
Desc Produk	Displayed in product detail	Product description	✅ Complete	Low	Low
Foto produk	Displayed via ImageGallery	Product photos	✅ Complete	Low	Low
Filter Produk	Only sorting exists	Comprehensive product filters	🟡 Partial	High	Medium
Spesifikasi Produk	Displayed in product tabs	Product specifications	✅ Complete	Low	Low
Inquiry Detail	Request for Quote feature	Inquiry detail form	✅ Complete	Low	Low

Portal Berita	Implemented as articles	News portal	✅ Complete	Low	Low
Akses Admin Portal Berita	CMS articles available	Admin access for news	✅ Complete	Low	Low
Perbandingan Produk	Implemented via compare	Compare products	✅ Complete	Low	Low
About Us	Exists as static page	About Us page	✅ Complete	Low	Low
Sistem Plts 3 Page	Exists as subcategories only	3 distinct pages for PLTS	🟡 Partial	Medium	Medium
Contact Details	Inside About Us page	Contact Details	🟡 Partial	Medium	Low
Promo Section	Promo page & Banners	Promo section	✅ Complete	Low	Low
Whatsapp Button	Floating WhatsApp used	WhatsApp button	✅ Complete	Low	Low
Penarikan Data customer	Saved to QuoteRequest	Leads data capture	✅ Complete	Low	Low

Feature Details

1. Landing Page

Current Status: ✅ Complete

- **Desired Behavior:** A comprehensive landing page for the website.
- **Current Implementation:** Implemented dynamically via `app/page.tsx` using `HomepageSection` database entries for flexible layout (Hero, Featured Products, Categories).
- **Evidence:** `src/app/page.tsx` , `HomepageSection` schema, `src/components/home/` .
- **Gap:** None.

2. Cart (Keranjang)

Current Status: ✅ Complete

- **Desired Behavior:** A functional shopping cart for users to accumulate items.
- **Current Implementation:** Fully implemented with state managed via `CartProvider` . Includes subtotal calculation and empty state handling.
- **Evidence:** `src/app/keranjang/page.tsx` , `src/providers/CartProvider.tsx` .
- **Gap:** None.

3. All katalog Product

Current Status: ✅ Complete

- **Desired Behavior:** A page to view all available products.
- **Current Implementation:** Implemented with pagination and sorting.

- **Evidence:** `src/app/produk/page.tsx` , `getProductsPaginated` in API.
- **Gap:** None.

4. Katalog per masing masing (Category Catalogs)

Current Status:  Complete

- **Desired Behavior:** Ability to view products filtered by specific categories or brands.
- **Current Implementation:** Dynamic routes for categories and brands exist, filtering products accordingly.
- **Evidence:** `src/app/kategori/[slug]/page.tsx` , `src/app/brand/[slug]/page.tsx` .
- **Gap:** None.

5. Desc Produk

Current Status:  Complete

- **Desired Behavior:** Product detail pages should have a description section.
- **Current Implementation:** Rendered correctly inside the Tabs component on the product detail page.
- **Evidence:** `src/app/produk/[slug]/page.tsx` , `description` field in `Product` schema.
- **Gap:** None.

6. Foto produk

Current Status:  Complete

- **Desired Behavior:** Display product images on product pages.
- **Current Implementation:** Fully implemented using a dynamic `ImageGallery` component handling multiple images.
- **Evidence:** `src/components/product/ImageGallery.tsx` , `images` JSON array in `Product` schema.
- **Gap:** None.

7. Filter Produk

Current Status:  Partial /  Needs UX Improvement

- **Desired Behavior:** Ability to filter products by various attributes (price, brand, specifications).
- **Current Implementation:** Currently, only a basic `SortDropdown` exists on catalog pages. There is no faceted sidebar for filtering by brand, price range, or categories simultaneously.
- **Evidence:** `src/app/produk/page.tsx` uses `<SortDropdown />` but lacks filter components.
- **Gap:** Missing comprehensive filtering UI.
- **Implementation Recommendation:** Build a sidebar filter component allowing multi-select for brands, categories, and price ranges.
- **Estimated Complexity:** Medium
- **Regression Risk:** Low

8. Spesifikasi Produk

Current Status:  Complete

- **Desired Behavior:** Product specifications presented clearly.
- **Current Implementation:** Handled cleanly via a JSON structure in the database and rendered in a dedicated "Spesifikasi" tab using a table layout.

- **Evidence:** `src/app/produk/[slug]/page.tsx` (Tabs section), `specifications` field in `Product` schema.
- **Gap:** None.

9. Inquiry Detail (Permintaan Penawaran)

Current Status:  Complete

- **Desired Behavior:** Users can send inquiries/quote requests for specific items.
- **Current Implementation:** Supported via an inquiry form that saves to the database.
- **Evidence:** `src/app/permintaan-penawaran/page.tsx` , `QuoteRequest` database model.
- **Gap:** None.

10. Portal Berita

Current Status:  Complete

- **Desired Behavior:** A news/blog portal for the website.
- **Current Implementation:** Implemented as the "Artikel" section.
- **Evidence:** `src/app/artikel/page.tsx` , `Article` database model.
- **Gap:** None.

11. Akses Admin untuk Portal Berita

Current Status:  Complete

- **Desired Behavior:** CMS/Admin access to manage news articles.
- **Current Implementation:** Dedicated admin interface for creating and managing articles.
- **Evidence:** `src/app/admin/articles/*` .
- **Gap:** None.

12. Perbandingan Produk dengan Produk lain

Current Status:  Complete

- **Desired Behavior:** Feature allowing users to compare multiple products.
- **Current Implementation:** Working comparison feature allowing users to toggle products for comparison.
- **Evidence:** `src/app/perbandingan/page.tsx` , `CompareToggleButton.tsx` .
- **Gap:** None.

13. About Us (Tentang Kami)

Current Status:  Complete

- **Desired Behavior:** "About Us" page detailing the company.
- **Current Implementation:** Available as a dynamically rendered static markdown page.
- **Evidence:** `src/lib/data/static-pages.ts` (slug: `tentang-kami`), `src/app/halaman/[slug]/page.tsx` .
- **Gap:** None.

14. Sistem PLTS 3 Page (On grid, Off Grid, Hybrid)

Current Status:  Partial /  Needs UX Improvement

- **Desired Behavior:** 3 distinct pages detailing "On grid", "Off Grid", and "Hybrid" PLTS systems.

- **Current Implementation:** Currently these are only implemented as product subcategories (e.g., `subcat-pkt-ongrid`), rather than dedicated informative pages explaining the difference between the systems.
- **Evidence:** `src/lib/data/categories.ts` .
- **Gap:** Missing dedicated, content-rich landing pages explaining how each system works.
- **Implementation Recommendation:** Create new routes (e.g., `/sistem/on-grid` , `/sistem/off-grid`) with detailed educational content and embedded related products.
- **Estimated Complexity:** Medium
- **Regression Risk:** Low

15. Contact Details

Current Status: 🟡 Partial / 🟡 Needs UX Improvement

- **Desired Behavior:** Clear contact details section/page for customers.
- **Current Implementation:** Contact information is currently bundled at the bottom of the "Tentang Kami" static page and likely in the footer.
- **Evidence:** `src/lib/data/static-pages.ts` .
- **Gap:** Lacks a dedicated `/kontak` page with an interactive map and direct contact form.
- **Implementation Recommendation:** Create a dedicated `app/kontak/page.tsx` page.
- **Estimated Complexity:** Low
- **Regression Risk:** Low

16. Promo Section

Current Status: ✅ Complete

- **Desired Behavior:** Section to highlight current promotions.
- **Current Implementation:** Dedicated promo route and database models for banners.
- **Evidence:** `src/app/promo/page.tsx` , `Banner` database model.
- **Gap:** None.

17. Whatsapp Button

Current Status: ✅ Complete

- **Desired Behavior:** Floating or inline WhatsApp button for quick contact.
- **Current Implementation:** Implemented via a floating component and inline buttons in the product detail pages.
- **Evidence:** `src/components/layout/FloatingWhatsApp.tsx` , `src/app/produk/[slug]/page.tsx` .
- **Gap:** None.

18. Penarikan Data untuk customer yang menghubungi

Current Status: ✅ Complete

- **Desired Behavior:** Ability to extract/view data of customers who contacted the business.
 - **Current Implementation:** Handled robustly by saving all inquiries to the `QuoteRequest` table, which is accessible via the admin dashboard.
 - **Evidence:** `QuoteRequest` model in `prisma/schema.prisma` , `src/app/admin/quotes/page.tsx` .
 - **Gap:** None.
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Implementation Roadmap

Phase 1: Critical Regressions

- *None detected.* The current system is highly functional and aligns well with the baseline requirements.

Phase 2: Missing Core Business Features

- *None missing.* All baseline requested functionality is at least partially present.

Phase 3: CMS Improvements

- **Sistem PLTS 3 Page:** Build a new CMS capability or static route set for dedicated informative pages (On-Grid, Off-Grid, Hybrid) combining markdown content with product carousels.

Phase 4: UX Improvements

- **Filter Produk:** Build a faceted sidebar filter for the `/produk` and `/kategori/[slug]` pages.
- **Dedicated Contact Page:** Separate contact information from the "About Us" markdown into a rich, dedicated `/kontak` page.

Phase 5: Nice-to-have Enhancements

- Allow direct CSV/Excel exports from the Admin Quotes section for easier offline CRM tracking.